

Candidate Generation After constructing the premise pool, we expand each premise into 4 story candidates using multiple LLMs. To ensure diversity, we select four representative models, including GPT-4o (OpenAI, 2024), Gemini 2.5 Pro (Comanici, 2025), Grok 4 (xAI, 2025), and Qwen-Long (Wan et al., 2025a). Each premise is provided to all four models under a unified prompting template, resulting in four candidate stories for each premise. These stories span a wide range of themes, languages, and narrative styles, providing a robust foundation for subsequent preference annotation. The construction details are in Appendix B.

2.2 PREFERENCE JUDGMENT

After collecting four story candidates for each premise, we rank them by quality, i.e., preference annotation. We first use several LLMs to score each LLM-generated story and rank the candidates accordingly. We then adopt human verification. To further identify which preference dimensions are most significant, e.g., whether *Story A* is preferred over *Story B* due to better coherence, we perform dimensional categorization.

Preference Ranking We adopt LLMs to score each LLM-generated story and rank the candidates accordingly. We then conduct human verification for the preference rank. Following prior work (Yang & Jin, 2024), we define five scoring dimensions: creativity, coherence, fluency, characterization, and relevance. We adopt four LLMs: GLM-4.5 (Team et al., 2025), Qwen2.5-Max (Yang et al., 2025), Claude-3.5 (Claude, 2024), and DeepSeek-R1 (DeepSeek-AI et al., 2025), to provide scores for each dimension and an overall score. Based on the **overall** scores, each LLM produces a ranking of the story candidates for each premise. We then compute the agreement among the rankings from four LLMs. Specifically, given a set of m candidate stories $\{c_1, c_2, \dots, c_m\}$ and the rankings produced by n LLMs, we measure the similarity between rankings using Kendall’s Tau correlation coefficient (Kendall, 1938). For any two models A and B , with ranking results denoted as π_A and π_B , respectively, Kendall’s Tau is defined as:

$$\tau(\pi_A, \pi_B) = \frac{N_c - N_d}{\binom{m}{2}}, \tau \in [-1, 1] \quad (1)$$

Let N_c denote the number of concordant pairs, N_d the number of discordant pairs, and $\binom{m}{2}$ the total number of possible story pairs. To assess the overall agreement among n LLMs, we compute Kendall’s Tau for every pair of models and take the average:

$$\tau_{\text{avg}} = \frac{2}{n(n-1)} \sum_{i < j} \tau(\pi_i, \pi_j) \quad (2)$$

If $\tau_{\text{avg}} < 0.6$, we consider it as strong disagreement and require human annotators to annotate the preference ranking. Otherwise, we use majority voting to obtain the final ranking, which is then human-verified. Additional annotation details are provided in Appendix C.

Dimensional Categorization After the previous section, we have obtained the preference rankings based on the overall scores. Since the evaluation considers five dimensions, we further perform dimensional categorization to identify which preference dimensions are most significant. For example, whether *Story A* is preferred over *Story B* due to superior coherence. Following the method of prior work (Yang & Jin, 2024), we design a four-stage method to categorize each prompt by determining which dimension is most decisive in distinguishing the preferred story from the rejected candidates. The full procedure is described in Appendix F. Our final dataset contains 1,133 preference instances. Based on the primary distinguishing dimension, the dataset is categorized into five groups: creativity (292), coherence (265), fluency (225), characterization (215), and relevance (136). The procedure of this method is as follows:

Stage 1: Mean Gap Analysis. For each dimension $d \in \{1, 2, \dots, 5\}$, we calculate the mean gap:

$$\text{mean gap}_d = \text{score}_{\text{chosen}, d} - \frac{1}{|R|} \sum_{r \in R} \text{score}_{r, d} \quad (3)$$

where R denotes the set of rejected stories. If one dimension exhibits a substantially larger mean gap than others (exceeding a predefined tie tolerance $\tau = 0.5$), we categorize the prompt group into that dimension.

Stage 2: Margin to Runner-up. When multiple dimensions show similar mean gaps, we examine the competitive margin:

$$\text{margin}_d = \text{score}_{\text{chosen},d} - \max_{r \in R} \text{score}_{r,d} \quad (4)$$

This identifies the dimension where the chosen story has the clearest separation from the best rejected alternative. We select the dimension with the largest margin.

Stage 3: Rejected Variance. If margins are tied, we consider the variance of rejected scores:

$$\text{variance}_d = \text{Var}(\{\text{score}_{r,d} : r \in R\}) \quad (5)$$

We select the dimension with the smallest variance, indicating consistent poor performance among rejected stories on this dimension.

Stage 4: Ordinal Fallback. If ties persist across all stages, we apply deterministic ordinal ranking ($d_1 \rightarrow d_2 \rightarrow \dots \rightarrow d_5$) to ensure categorical assignment.

3 STORYREWARD: A REWARD MODEL FOR STORY PREFERENCE

This section introduces the development of STORYREWARD, including the construction of large-scale high-quality story preference pairs (§ 3.1) and the training of STORYREWARD (§ 3.2).

3.1 COLLECTING PREFERENCE PAIRS

To ensure real-world coverage of the collected preference pairs and reduce human annotation costs, we design an automated data collection framework to gather human-written stories and various human preferences. Specifically, we first crawl a large number of human-authored stories from popular Chinese (Douban²) and English (WritingPrompts³) online literary platforms, along with associated metadata such as categories and upvote counts. We then design three methods to automatically collect preference pairs.

Premise Back-generation We collect a large number of stories from the Douban website. To collect pairwise data, we first cluster stories based on similarity, where stories within the same cluster either share platform-defined genre labels or originate from the same author’s column. Such clusters typically exhibit similarity in theme, writing style, and subject matter. Within each cluster, we construct story pairs by sampling two different stories and providing them as input to GPT-4o (OpenAI, 2024), which generates a premise based on the commonalities of the two stories. For preference judgment, we leverage user engagement statistics: specifically, the story with a higher readership count is considered as the “chosen”. To ensure reliability, we filter out pairs with extremely low engagement, ultimately yielding approximately 6,000 preference pairs derived entirely from authentic human-written stories. More details are provided in Appendix E.2.

Prompt-Guided Rewriting We select a human-written story and its corresponding premise as the “chosen” example. We then generate the “rejected” counterpart by prompting an LLM to rewrite the original story. Crucially, this rewriting process is not arbitrary. We employ a set of carefully designed prompt templates to systematically control the modifications and target specific aspects of narrative quality. These prompts constrain the LLM’s changes. For instance, by instructing it to alter only the story’s ending, it can undermine plot coherence. Further details are provided in Appendix E.3. This constrained approach prevents trivial

²<https://www.douban.com>

³<https://huggingface.co/datasets/euclaise/writingprompts>

discrepancies between the chosen and rejected stories. Instead, it produces rejected samples that remain readable but are deliberately flawed in their style, tone, or logic. This method is based on the assumption that human-written stories are generally of higher quality, an assumption we also validate on 20 samples.

Human-Guided Continuation In preliminary experiments, we observe that directly contrasting human-written stories with LLM-generated ones during reward model training often leads to rapid convergence and suboptimal performance. This is primarily due to the substantial quality gap and surface differences between human-written and LLM-generated stories. To mitigate this issue, we provide two LLMs with identical premises and prompts, and only one LLM can access the genuine human continuation of the story. After generating stories from both settings, we construct preference pairs under the hypothesis that the model guided by human-authored continuations is more likely to produce superior narratives. To validate this hypothesis, we randomly sample 20 preference pairs and determine that the model guided by human-authored continuations performs better. More generation details and prompts are provided in Appendix E.4.

In addition to constructing story preference pairs from human-written stories, we also adopt a widely used approach (Cui et al., 2024), where an LLM generates two stories for the same premise and another LLM automatically labels one as chosen and the other as rejected. More details on generation settings and prompts are provided in Appendix E.1. In total, we obtain about 100,000 high-quality preference pairs that capture diverse and real-world human preferences, which can be directly used for reward model training.

3.2 TRAINING STORYREWARD

We leverage Llama-3.1-8B-Instruct and Qwen3-8B as the base models for training. We use the premise of each instance in our collected preference pairs as the input and the stories as the preference pairs to train STORYREWARD-LLAMA and STORYREWARD-QWEN. We set the batch size to 1, the learning rate to 9×10^{-6} for STORYREWARD-LLAMA and 2×10^{-5} for STORYREWARD-QWEN, and train for 1 epoch.

4 EXPERIMENTS

4.1 MAIN EXPERIMENTS

Experimental Setup Formally, given a set of four candidate stories $\{s_1, s_2, s_3, s_4\}$, the RMs assign a score $f(s_i)$ to each candidate. A prediction is considered correct if $\arg \max_i f(s_i) = s^*$, where s^* denotes the predicted chosen story. We report the proportion of correct predictions as the primary metric for evaluation. We categorize the baselines into two main groups. (1) Reward models, which are explicitly trained for reward modeling, typically formulated as regression models that assign a score to each response and select the one with the highest score. We include several advanced and representative models in this category, such as InternLM2-Reward (Cai et al., 2024), Skywork-Reward (Liu et al., 2024), ArmoRM (Wang et al., 2024a), QRM (Dorka, 2024), GRM (Yang et al., 2024), and WQRM (Chakrabarty et al., 2025). (2) LLM-based generative reward models, where large language models are employed to either directly score responses or conduct pairwise comparisons to select the preferred response (Lambert et al., 2024). In this group, we evaluate several proprietary systems, including GPT-4o (OpenAI, 2024), Gemini-2.5 Pro (Comanici, 2025), Grok (xAI, 2025), our base model Llama-3.1-8B Instruct (Meta, 2024b), and Qwen3-8B (Yang et al., 2025).

Experimental Results All the experimental results are presented in Table 1. We observe the following: (1) Existing reward models and LLM-as-judges perform poorly. The best model GPT-4o achieves only 66.3%, which is far from applicable in downstream applications. This indicates current reward models neglect the modeling of story preferences and further demonstrates the complexity of story preference, which requires dedicated data and modeling rather than relying on general-domain preference generalization. (2) In general, reward models perform better on the *fluency* dimension, tending to favor more fluent stories. This